GULU UNIVERSITY

Version 1.0 | September 2026

Derived from the Gulu University 2026 RIF documents and the approved business-process blueprint

Purpose: Functional blueprint for UI/UX, database, backend/API, testing and implementation

Document Control

Item	Specification
Document	Gulu RIF-MIS System Requirements & Module Specification
Version	1.0
System name	Gulu University Research & Innovation Fund Management Information System (Gulu RIF-MIS)
Primary institutional owner	Directorate of Research and Graduate Training (DRGT)
Decision authority referenced in source form	Board of Research and Graduate Training
Primary source	Gulu University 2026 RIF Call + 2026 RIF Application Form/Detailed Proposal Guidelines
Related blueprint	Gulu RIF-MIS Detailed Business Process Specification v1.0
Status	Proposed system specification for institutional validation

1. System Vision

Gulu RIF-MIS is proposed as a secure institutional platform integrated with the Gulu University website and research-management environment. It will digitize the RIF lifecycle from call publication and application through eligibility screening, technical review, decision, award, implementation, reporting, monitoring, close-out and impact tracking.

The system is not intended to replace Gulu University's RIF policy. It is a workflow and information-management layer that operationalizes approved policy and procedures.

2. Requirements Basis

The Gulu 2026 RIF documentation already defines the major application information areas: applicant/team, project information, priority areas, duration and budget, executive summary, problem statement, objectives, questions, outputs, outcomes, impacts, methodology, knowledge translation, innovation, commercialization, industry links, inclusion, ethics, risk, workplan and budget. The official-use section also includes eligibility, technical review score, funding decision, amount awarded and comments. filecite turn0file0 L10-L24 filecite turn0file0 L25-L37 filecite turn0file0 L41-L59 filecite turn0file0 L60-L120 filecite turn0file0 L121-L191 filecite turn0file0 L198-L249

The call also defines eligibility, funding categories, deliverables, application requirements, evaluation areas and submission dates. filecite turn0file1 L74-L110 filecite turn0file1 L113-L139

3. High-Level System Architecture

Recommended logical structure:

PUBLIC GULU UNIVERSITY WEBSITE → RIF PUBLIC PORTAL → AUTHENTICATION → ROLE-BASED WORKSPACES → APPLICATION ENGINE → ELIGIBILITY ENGINE → REVIEW ENGINE → DECISION ENGINE → AWARD/PROJECT ENGINE → REPORTING & M&E → INNOVATION/IP/COMMERCIALIZATION → CLOSE-OUT & IMPACT

Layer	Main responsibility
Presentation	Public portal, applicant portal, reviewer portal, DRGT dashboard, committee dashboard, project dashboard.
Application/API	Authentication, workflow, validation, notifications, business rules, reporting APIs.
Domain services	Calls, applications, reviews, awards, projects, finance

	status, M&E, innovation and audit.
Data	Relational database for structured records and secure document storage.
Integration	University website, identity/email, finance and repository integrations where approved.

4. Module Catalogue

ID	Module	Primary users	Priority
M01	Public RIF Portal	Public	MVP
M02	User, Identity & Role Management	All / Admin	MVP
M03	Researcher Profile & Team Management	Applicant/PI	MVP
M04	RIF Cycle & Call Management	DRGT	MVP
M05	Application Management	Applicant/PI	MVP
M06	Eligibility & Administrative Screening	DRGT	MVP
M07	Reviewer Management	DRGT/Admin	MVP
M08	Technical Review & Scoring	Reviewer	MVP
M09	Board/Committee Decision Management	Board/Committee/DRGT	MVP
M10	Award & Contract Management	DRGT/PI	MVP
M11	Project & Milestone Management	PI/DRGT/M&E	MVP
M12	Financial/Budget Monitoring	Finance/DRGT/PI	Phase 2
M13	Reporting & Accountability	PI/DRGT/Finance	Phase 2
M14	Monitoring & Evaluation	M&E/DRGT	Phase 2
M15	Innovation, IP & Commercialization	Innovation/DRGT	Phase 2
M16	Knowledge Translation & Impact	DRGT/PI/M&E	Phase 2
M17	Notifications & Communication	All authorized users	MVP
M18	Dashboards & Analytics	Management/DRGT/Board	MVP
M19	Document & Records Management	All authorized users	MVP
M20	Audit, Compliance & System Administration	Admin/DRGT	MVP

5. M01 — Public RIF Portal

Purpose: Provide the public-facing entry point for RIF information and integrate with the University's existing website.

Function	Requirement
Calls	Display active and archived RIF calls.
Call details	Show theme, priorities, categories, eligibility, requirements and dates.
Guidance	Publish application instructions, templates and FAQs.
Status	Show whether a call is open, closed or under review

	where appropriate.
Announcements	Publish official RIF notices.
Results	Publish approved/public information only after authorized release.
Entry point	Link users to secure applicant login/application system.

The current 2026 call identifies the official theme, objectives, priority areas, eligibility, funding categories, deliverables, requirements and dates; these should be represented as configurable call content rather than hard-coded website pages. filecite turn0file1 L3-L18 filecite turn0file1 L19-L73

6. M02 — User, Identity & Role Management

- User registration/authentication.
- University identity integration where available.
- Role assignment and revocation.
- Account activation/deactivation.
- Password/session controls if local authentication is used.
- Role-based permissions.
- Department/faculty/unit association.
- Audit of privileged account changes.

Role	Key permissions
Applicant/PI	Create/manage applications; submit; view decisions; manage project reports.
Co-investigator	Contribute to permitted application/project sections.
DRGT Administrator	Manage calls, screening, workflow, reviews and records.
Reviewer	Access assigned proposals and submit reviews.
Board/Committee Member	Access decision docket and record authorized decision input.
Finance	Access financial functions assigned by policy.
M&E	Access monitoring/reporting functions.
Innovation Officer	Access innovation/IP/commercialization functions.
System Administrator	Technical configuration and access administration.

7. M03 — Researcher Profile & Team Management

Function	Requirement
Profile	Maintain reusable researcher information.
Team	Create project team and assign roles.
Invitation	Invite co-investigators/team members.
Collaboration	Capture external collaborators where permitted.
Declarations	Collect required declarations/confirmations.
History	Retain application-specific team snapshots so later profile edits do not rewrite historical applications.

8. M04 — RIF Cycle & Call Management

This is the administrative control centre for creating a funding cycle.

Feature	System requirement
Create cycle	Create financial year/cycle and unique reference.
Theme	Store official theme.
Priorities	Configure approved priority research areas.

Categories	Configure funding categories.
Eligibility	Configure screening checklist/rules.
Requirements	Configure mandatory sections and documents.
Dates	Configure opening, deadline, review, announcement and project-start dates.
Evaluation	Configure criteria and weights.
Workflow	Configure status transitions and responsible roles.
Publication	Submit call for internal approval, then publish.
Close	Automatically or manually close submissions at deadline.

The 2026 call's dates include opening 9 September 2026, deadline 8 October 2026, review 12–30 October 2026, announcement 13 November 2026 and project commencement 20 November 2026. These should be stored as call configuration, not hard-coded system constants. filecite turn0file1 L130-L139

9. M05 — Application Management

The application engine is the core applicant-facing module.

Section	System functions
Applicant/team	PI, co-investigators, external collaborators.
Project information	Title, category, priority, duration, requested budget.
Executive summary	Structured text field with configured word guidance.
Problem & justification	Structured proposal sections.
Objectives/questions	Structured capture.
Outputs/outcomes/impact	Structured result capture.
Methodology	Research/innovation approach and data methods.
Knowledge translation	Dissemination and policy engagement.
Innovation	Innovation component and commercialization.
Industry/technology transfer	Partners and transfer plan.
Inclusion	Gender, youth, vulnerable populations and inclusivity.
Ethics/risk	Ethics plan and project risks.
Workplan	Activities, dates and outputs.
Budget	Itemized budget and justification.
Declarations	Submission declaration and required confirmations.
Attachments	CVs, letters, ethics documents, instruments and other required files.

The detailed guidelines require a 250–500 word executive summary and specify sections including background, problem statement, justification, objectives, literature review, framework/theory of change, methodology, innovation, policy relevance, commercialization/industrial linkages, inclusion, ethics/risk, results/impact pathway, knowledge translation, sustainability, work plan, results framework, budget, references and appendices. filecite turn0file0 L278-L349

10. M06 — Eligibility & Administrative Screening

Feature	Requirement
Queue	Display submitted applications awaiting screening.
Checklist	Configurable eligibility criteria.
Evidence	Reviewer/admin records evidence or comment.
Decision	Eligible / Ineligible / Other approved status.
Reason	Required explanation for negative status.
Correction	Only if institutionally approved; configurable.
Lock	Prevent unauthorized edits after screening/review begins.

Audit	Record who screened, when and what was decided.

11. M07 — Reviewer Management

- Reviewer directory and expertise profile.
- Reviewer availability.
- Assignment by authorized DRGT users.
- Conflict-of-interest declaration.
- Reviewer acceptance/decline.
- Deadline tracking.
- Reminder/escalation.
- Reviewer workload dashboard.
- Reviewer history and audit record.
- Controlled replacement of reviewers.

The exact number of reviewers per proposal is intentionally not fixed here because the supplied Gulu documents do not specify it.

12. M08 — Technical Review & Scoring

The review module shall implement Gulu's published detailed evaluation logic as configurable criteria.

Criterion	Weight
Scientific rigor	25%
Innovation potential	20%
Policy relevance	15%
Impact potential	20%
Feasibility	10%
Team strength	5%
Budget efficiency	5%
TOTAL	100%

Source: Gulu University's 2026 Detailed Proposal Guidelines. filecite turn0file0 L350-L361

- Reviewer opens assigned proposal.
- Reviewer confirms conflict-of-interest status.
- Reviewer scores each criterion.
- Reviewer enters comments/justification.
- System calculates weighted score.
- Reviewer submits review.
- Submitted review becomes locked.
- DRGT sees review completion status.
- Authorized decision users see aggregated review information.

13. M09 — Board/Committee Decision Management

Feature	Requirement
Decision docket	List review-ready applications.
Application summary	Proposal, team, eligibility, scores, comments and budget.
Decision	Configurable decision options approved by University.

Approved amount	Separate from requested amount.
Comments	Record official decision comments.
Approval	Authorized chair/official confirmation.
Notification	Trigger applicant notification after authorized release.
Audit	Immutable decision history.

The application form's official-use section explicitly captures eligibility, technical review score, funding decision, amount awarded and comments, with the Chairperson of the Board of Research and Graduate Training shown in the decision section. filecite turn0file0 L213-L249

14. M10 — Award & Contract Management

- Convert approved application into award.
- Store approved amount and project dates.
- Store approved workplan/milestones.
- Store award conditions.
- Upload/generate agreement documents.
- Track acceptance/signature.
- Track prerequisites before activation.
- Maintain award amendments and versions.
- Link award to project and financial records.

15. M11 — Project & Milestone Management

Feature	Requirement
Project profile	Approved title, PI, team, amount, dates and category.
Milestones	Due dates, outputs, responsible persons and status.
Activities	Detailed implementation activities.
Risks/issues	Register and track risks/issues.
Evidence	Attach evidence to milestones and outputs.
Changes	Controlled variation/extension requests.
Completion	Track completion against approved plan.

16. M12 — Financial/Budget Monitoring

- Approved budget baseline.
- Budget-line monitoring.
- Requested vs approved amount.
- Release/disbursement status.
- Expenditure/accountability status.
- Remaining balance.
- Budget variation requests.
- Finance notes and evidence.
- Integration-ready API rather than duplicate accounting ledger.

The RIF application form already requires a work plan and budget, while the guidelines require detailed budget and justification. filecite turn0file0 L198-L212 filecite turn0file0 L337-L341

17. M13 — Reporting & Accountability

Report type	System capability
Progress report	Online form, evidence upload, submission and review.
Financial accountability	Financial data/evidence according to approved University process.
Final report	Final outputs, outcomes, impact, lessons and evidence.
Return for correction	Reviewer returns report with comments where authorized.
Approval	Authorized office accepts/finalizes report.
Compliance	Dashboard of due, submitted, overdue and accepted reports.

18. M14 — Monitoring & Evaluation

- Portfolio monitoring dashboard.
- Project-level milestone monitoring.
- Risk and issue tracking.
- Field/desk monitoring records.
- Indicator tracking.
- Outcome and impact evidence.
- Corrective action tracking.
- Monitoring history and audit trail.

Gulu's guidelines require an impact pathway/results framework, knowledge translation and sustainability considerations, making M&E a core rather than optional component. filecite turn0file0 L325-L341

19. M15 — Innovation, IP & Commercialization

- Identify projects with innovation components.
- Track innovation/prototype/technology records.
- Record industry partners.
- Track validation and technology-transfer actions.
- Track IP-related actions and documentation.
- Track commercialization milestones.
- Record uptake/adoption evidence.
- Link innovation records to project outputs.

The Gulu guidelines specifically request innovation, commercialization potential, industry partners and technology-transfer plans. filecite turn0file0 L143-L171

20. M16 — Knowledge Translation & Impact

Area	Examples
Knowledge outputs	Journal articles, books, reports, policy briefs.
Policy engagement	Stakeholder meetings, policy engagement and evidence uptake.
Community	Beneficiaries, community products and practical uptake.
Industry	Partnerships, adoption, commercialization.
Institutional	Capacity building and institutional transformation.

Impact	Economic, social, policy, institutional and environmental evidence.

These categories reflect the outputs, outcomes and impact areas in the Gulu application form. fileciteturn0file0 L91-L120

21. M17 — Notifications & Communication

Trigger	Channel
Submission confirmation	Email + in-system notification
Screening result	Email + portal
Review assignment	Email + portal
Review reminder	Email + portal
Decision	Email + portal
Award action required	Email + portal
Report due	Email + portal
Report overdue	Email + portal
Variation decision	Email + portal
Close-out action	Email + portal

Email content should use official DRGT communication templates and should not disclose confidential reviewer information.

22. M18 — Dashboards & Analytics

Dashboard	Widgets / data
Public	Open calls, deadlines, published guidance and approved public results.
DRGT	Applications by stage, screening backlog, review completion, decisions, active projects, overdue reports.
Board	Decision docket, review summaries, requested/approved amounts and strategic portfolio information.
Applicant	My applications, statuses, actions required, projects, reports and notifications.
Reviewer	Assigned proposals, deadlines and completion.
M&E	Projects at risk, milestones, reports and impact indicators.
Finance	Portfolio financial status and budget utilization where integrated.
Innovation	Innovations, prototypes, IP actions, industry partnerships and commercialization.

23. M19 — Document & Records Management

- Secure file upload/download.
- File-type and size validation.
- Version control for editable records.
- Association with application, review, award, project or report.
- Access control by role.
- Document metadata.
- Retention/archive controls.
- Audit of upload, replacement and access actions.

24. M20 — Audit, Compliance & Administration

Administrative area	Requirement
Audit trail	Record key create/update/submit/approve/reject actions.
Configuration	Manage call rules, criteria, templates and workflow settings.
Roles	Assign/revoke permissions.
Reference data	Manage faculties, departments, categories and statuses.
System logs	Monitor application and integration errors.
Backup	Scheduled backup and tested recovery.
Compliance	Support institutional audits and record retrieval.

25. Cross-Module Workflow

M04 Call → M05 Application → M06 Screening → M07 Reviewer Assignment → M08 Review → M09 Decision → M10 Award → M11 Project → M12 Financial + M13 Reporting + M14 M&E → M15 Innovation/M16 Impact → Close-out → Portfolio Analytics

No module should create a parallel source of truth for the same business object. For example, the approved project budget should originate from the award/project record and be exposed to finance/reporting rather than manually recreated in several modules.

26. Core Database Model — Initial

Entity	Key relationships
users	roles, researcher_profile, audit_events
researcher_profiles	user, team_memberships
rif_cycles	rif_calls
rif_calls	priority_areas, funding_categories, applications
applications	applicant, call, team, sections, budget, attachments, screening, reviews
application_versions	application
teams/team_members	application, researcher
proposal_sections	application
workplans/milestones	application/project
budgets/budget_lines	application/award/project
attachments	application/review/report/award/project
eligibility_checks	application
review_assignments	application/reviewer
reviews/review_scores	review_assignment
decisions	application
awards/contracts	application/project
projects	award, team, milestones, reports
reports	project
monitoring_records	project
risks/issues	project
change_requests	project/award
innovations	project
ip_records	innovation/project
commercialization_activities	innovation/project
notifications	user/application/project
audit_events	all major entities

27. API / Backend Service Requirements

Service group	Example operations
Authentication	login, logout, session/token, role check
Calls	create, configure, publish, close, archive
Applications	create, update, validate, submit, retrieve
Screening	assign, checklist, decide, return
Reviews	assign, accept, score, submit, lock
Decisions	prepare docket, record decision, approve/release
Awards	create, activate, amend, close
Projects	create, milestones, risks, changes, complete
Reports	create, submit, review, accept
Finance	budget status, releases, expenditure/status integration
M&E	indicators, monitoring, actions
Innovation	innovation, IP, commercialization
Notifications	templates, send, reminders
Analytics	dashboard aggregates and exports
Audit	event logging and retrieval

28. Validation Rules

- Required fields cannot be blank.
- Application cannot be submitted after the configured deadline unless an authorized override exists.
- Budget totals must reconcile with line items.
- Workplan dates must fall within the project period unless approved otherwise.
- Required documents must be present before submission.
- Only authorized users may change submitted applications.
- Reviewer cannot submit a review without required scores/comments.
- Decision amount cannot exceed applicable approved limits unless an authorized exception rule exists.
- Project cannot become active until required award conditions are complete.
- Reports cannot be finalized by unauthorized users.

29. Security Requirements

- Role-based access control at both UI and API levels.
- Server-side authorization; never rely only on hidden UI controls.
- Secure password hashing if local authentication is used.
- Session timeout and account protection.
- HTTPS in production.
- Input validation and output encoding.
- File upload scanning/validation.
- Protection against common web vulnerabilities.
- Audit logging for privileged actions.
- Restricted access to reviewer identities/comments.
- Regular backups and recovery testing.

30. Suggested Technical Stack

This section is a proposed implementation option, not a Gulu University policy requirement.

Layer	Recommended option
Frontend	HTML5, Bootstrap 5, JavaScript; optionally a modern component framework if required.
Backend	Python + Flask, using a modular service architecture.
API	RESTful JSON API with documented endpoints.
Database	MariaDB/MySQL-compatible relational database.
ORM/data access	SQLAlchemy or controlled repository/service layer.
Background jobs	Celery/RQ or equivalent for notifications and scheduled tasks.
Document storage	Secure filesystem/object storage with database metadata.
Authentication	University SSO/identity integration where available; otherwise secure local identity.
Charts	ApexCharts/Chart.js or equivalent.
Deployment	University-controlled Linux server/cloud infrastructure subject to IT policy.
Version control	Git-based repository and controlled deployment pipeline.

31. MVP Boundary

The first production-capable release should concentrate on the workflow that solves the largest current administrative gap: moving from call + emailed documents to a traceable online funding workflow.

- M01 Public Portal
- M02 Identity/Roles
- M03 Researcher/Team
- M04 Call Management
- M05 Application
- M06 Eligibility
- M07 Reviewer Management
- M08 Technical Review
- M09 Decision
- M10 Award
- M11 Basic Project/Milestone
- M17 Notifications
- M18 Core Dashboards
- M19 Documents
- M20 Audit/Admin

Financial integration, advanced M&E, innovation/IP/commercialization and advanced analytics can follow after the core workflow is stable, while still designing the database and APIs so these modules can be added without redesigning the foundation.

32. Phase 2 Modules

- Financial integration and detailed accountability.
- Advanced M&E and impact indicators.
- Innovation/IP/commercialization tracking.
- Knowledge translation and portfolio impact analytics.
- Research repository integration.

- Advanced management reports and exports.
- External partner/collaborator features where approved.

33. Testing Requirements

Test area	Minimum coverage
Authentication	Login, logout, role restrictions, account recovery.
Application	Save, edit, validation, attachments, submission and deadline handling.
Screening	Checklist, decisions, audit.
Review	Assignment, conflict, scoring, locking, reminders.
Decision	Docket, decision, approved amount, authorization.
Award	Activation, conditions, amendments.
Project	Milestones, evidence, changes.
Reports	Submission, review, return, acceptance.
Security	Authorization, file access, input validation, audit.
Integration	Email, website links and approved external systems.
Performance	Expected concurrent users and document operations.
Recovery	Backup/restore and failure recovery.

34. Traceability Matrix — Source to System

Gulu source requirement	System module
Applicant and research team information	M03/M05
Project categories and priority areas	M04/M05
Proposal narrative and methodology	M05
Outputs, outcomes and impact	M05/M16
Knowledge translation/policy engagement	M05/M16
Innovation/commercialization/industry	M05/M15
Gender/youth/inclusion/ethics/risk	M05/M14
Workplan and budget	M05/M11/M12
Eligibility check	M06
Technical review score	M08
Funding decision and amount awarded	M09/M10
Reports and results framework	M13/M14
Call dates and submission process	M04/M17

35. Requirements Requiring Gulu University Approval

Before coding these rules as final business logic, the University should confirm:

- Exact eligibility criteria and exclusions.
- Reviewer count and reviewer qualification requirements.
- Conflict-of-interest rules.
- Score aggregation/moderation method.
- Decision and funding-allocation rules.
- Funding release and accountability workflow.
- Procurement requirements.
- Report frequency and templates.
- Variation/extension thresholds.
- Appeal/complaint process.

- IP ownership and commercialization rules.
- Record retention and public disclosure rules.
- Integration permissions with finance, identity and repository systems.

36. Recommended Development Sequence

1. Validate this specification with DRGT and relevant institutional stakeholders.
2. Convert requirements into UI wireframes and screen specifications.
3. Design the relational database and workflow/state model.
4. Design API contracts.
5. Build authentication and role management.
6. Build call/application workflow.
7. Build eligibility and review workflow.
8. Build decision/award workflow.
9. Build project/milestone/reporting foundation.
10. Build dashboards, audit and notifications.
11. Pilot with a controlled internal test call.
12. Conduct security, usability and workflow acceptance testing.
13. Train DRGT/admin/reviewer users.
14. Deploy under institutional IT controls.

37. Final System Concept

The resulting platform should behave as an institutional RIF workflow engine, not merely an online application form. An application should become a traceable record that moves through controlled states, with every review, decision, award, project milestone, report and close-out action connected to the same lifecycle.

This design preserves the substance of Gulu University's 2026 RIF process while providing the digital infrastructure needed for transparency, traceability, efficiency, portfolio monitoring and future expansion.

Appendix A — Proposed Navigation

Workspace	Main navigation
Public	Home | Open Calls | Guidelines | Announcements | Results | Login
Applicant	Dashboard | My Applications | New Application | My Projects | Reports | Notifications | Profile
DRGT	Dashboard | Calls | Applications | Screening | Reviews | Decisions | Awards | Projects | Reports | M&E | Innovation | Analytics
Reviewer	Dashboard | Assigned Proposals | Reviews | Deadlines | Profile/Conflict Declaration
Board/Committee	Dashboard | Decision Docket | Proposal Details | Review Summary | Decisions
Finance	Dashboard | Awards | Budgets | Financial Status | Accountabilities
M&E	Dashboard | Projects | Milestones | Reports | Risks | Indicators | Monitoring
Innovation	Dashboard | Innovations | IP | Industry Partners | Commercialization
Admin	Users | Roles | Reference Data | Workflow | Templates |

	Audit | System Settings

Appendix B — Core User Journey

Applicant: Login → Profile → Open Call → New Application → Complete Sections → Upload Documents → Validate → Submit → Track Status → Respond to Requests → Award → Manage Project → Submit Reports → Close-out.

DRGT: Create Call → Publish → Monitor Applications → Screen → Assign Reviewers → Monitor Reviews → Prepare Decision Docket → Record Decision → Manage Awards → Monitor Projects → Review Reports → Close-out → Portfolio Analytics.

Reviewer: Login → Conflict Declaration → Assigned Proposal → Review Criteria → Score + Comments → Submit → Lock.

Board/Committee: Login → Decision Docket → Proposal/Review Summary → Record Authorized Decision → Confirm.

M&E: Portfolio → Project → Milestones/Risks → Reports → Evidence → Monitoring Decision → Corrective Action → Impact.

Appendix C — Design Principle

Build the system around the RIF lifecycle and business objects, not around individual web pages. This allows the same application record to flow from submission to review, award, project, reporting and impact without duplicate data entry.